Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Yuhan Lee	**Email**	Ylee21@bwh.harvard.edu
Institute Name	Mass General Brigham (Harvard Medical School)			
Address	60 Fenwood Rd. Boston, MA 02115, USA			

2. Technology Overview

Title of Proposed Technology	Injectable Ultra-High-Concentration Protein Therapeutic Formulations Enabled by Particle-Based and Non-Aqueous Delivery Platforms			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Biotechnology / Pharmaceutical			
	Medium Category	Small Category	Medium Category	Small Category
	Biopharmaceutical formulation and delivery	*Injectable protein formulations, drug delivery systems*		
Concept of Proposed Technology	This technology aims to enable ultra-high-concentration (UHC) injectable formulation platforms of protein therapeutics, including antibodies, Fc-fusion proteins, enzymes, and other biologics, at concentrations exceeding those achievable with conventional aqueous formulations. The core concept is to decouple dose from solution viscosity and instability by formulating proteins into engineered solid or semi-solid states (e.g., microparticles, dispersions, suspensions, liquid-based nanoparticles) that can be injected subcutaneously using standard needles or autoinjectors. Rather than relying solely on highly concentrated aqueous solutions, the platform leverages: • Particle-based protein formulations (e.g., spray-dried or otherwise engineered protein particles) • Stabilizing excipients that protect proteins from aggregation and interfacial damage			

(max 10 pages)

	• Non-aqueous or low-water carrier systems that reduce protein–protein interactions • Injectability-optimized rheology to enable clinically acceptable injection forces This approach allows delivery of gram-scale protein doses in single or low-volume injections, potentially replacing intravenous infusions and enabling at-home administration.
Rationale for Joint R&D Project	Many clinically approved protein therapeutics require high doses but are limited by formulation viscosity, instability, and injection volume constraints, leading to intravenous administration or multiple injections per dose. These limitations increase patient burden, healthcare costs, and barriers to global access. While a few recent academic works have demonstrated feasibility of UHC protein suspensions using specific spray-dried particles, significant translational challenges remain, including: • Scale-up and manufacturing robustness • Generalizability across diverse protein modalities • Long-term stability and device compatibility • Regulatory-ready formulation strategies Collaboration with Korean SMEs is critical to address these gaps, particularly in: • Scalable particle manufacturing • Pharmaceutical excipient development • Injectable device integration • Process engineering and quality control This joint R&D project will bridge academic formulation innovation with industrial manufacturing and commercialization capabilities.
Key Words	1. Ultra-high-concentration protein formulation 2. Injectable biologics 3. Particle-based drug delivery 4. Subcutaneous administration 5. Biopharmaceutical manufacturing

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (TRL 1–3)	**End of the Project**	Level (TRL 5–6)
R&D Objectives	1. **Develop a broadly applicable injectable formulation platform** that enables ultra-high-concentration protein therapeutics while maintaining injectability, stability, and biological activity. 2. **Establish scalable and manufacturable formulation strategies** compatible with clinically relevant subcutaneous delivery formats, including standard syringes and autoinjectors. 3. **Demonstrate translational feasibility** through prototype formulations that reduce injection volume and administration burden without compromising pharmacokinetics or efficacy.			
R&D Target Specifications	1. Protein concentration: ≥300 mg/mL (target up to 600 mg/mL) 2. Injection force: ≤25-30 N through 21–27G needles 3. Injection volume: ≤3 mL per dose 4. Protein integrity: ≥70% monomer after formulation and storage 5. Storage stability: ≥3 months under controlled conditions			
R&D Plan	**Aim 1**. <u>Formulation Design Space Mapping and Candidate Identification</u> (Suggested timeline: Months 0–18) The objective of Aim 1 is to systematically define the design space for injectable ultra-high-concentration (UHC) protein therapeutics and to identify a limited set of candidate formulation systems for downstream development. This aim focuses on understanding fundamental constraints and opportunities, rather than committing to a single formulation approach at the outset. This aim will explore how protein properties, formulation architecture, excipients, and carrier environments interact to enable high effective concentrations while preserving injectability and stability. *Aim 1.1*: <u>Protein-dependent constraint mapping</u> Representative protein therapeutics, including monoclonal antibodies, Fc-fusion proteins, enzymes, and other biologics, will be evaluated to understand how molecular attributes (e.g., size, charge distribution, surface hydrophobicity, and conformational stability) influence aggregation, viscosity, and interfacial sensitivity at high effective concentrations. This analysis will distinguish constraints that are broadly shared across protein classes from those that are modality-specific. *Aim 1.2*: <u>Exploration of formulation architectures that decouple dose from solution concentration</u> Multiple formulation architectures will be explored that reduce reliance on highly concentrated aqueous solutions. These may include particle-based systems (e.g., spray-dried or otherwise engineered protein particles), as well as alternative solid, semi-solid, or hybrid formats. Emphasis will be placed on understanding how physical form and microstructure influence injectability, rheology, and stability. *Aim 1.3*: <u>Excipients and carrier environment screening</u> Stabilizing excipients and carrier environments will be screened to mitigate aggregation, interfacial denaturation, and stress-induced degradation. Both established pharmaceutical excipients and emerging material classes may be considered, along with aqueous, low-water, or non-aqueous carrier systems. This sub-aim will establish early design principles linking excipient chemistry and carrier properties to formulation performance. *Aim 1.4*: <u>Structure-property-performance integration and candidate selection</u> Data generated across Aim 1 will be integrated to establish preliminary structure-property-performance relationships connecting protein characteristics, formulation architecture, excipient			

selection, and macroscopic outcomes such as injectability and stability. Based on these insights, a small number of candidate formulation systems (defined by both material composition and physical form e.g., particle-based, suspension-based, or hybrid formulations) will be selected to represent distinct and promising design strategies.

Aim 2. Optimization of Injectability, Stability, and Handling Robustness (Suggested timeline: Months 12–30)

The objective of Aim 2 is to refine and optimize the candidate formulation systems identified in Aim 1 to ensure practical performance under clinically relevant conditions. This aim emphasizes injectability, robustness, and usability rather than final commercial specifications.

Aim 2.1: Injectability and rheological optimization
Candidate formulations will be evaluated for flow behavior, shear response, and injection force using clinically relevant needles and delivery conditions. Formulation parameters will be iteratively adjusted to ensure compatibility with standard syringes and autoinjector systems while avoiding excessive injection forces or flow instabilities.

Aim 2.2: Stability under stress and handling conditions
Formulations will be subjected to stress conditions such as temperature excursions, agitation, and freeze–thaw cycling to identify dominant degradation pathways. These studies will inform mitigation strategies and guide formulation refinement, without establishing final shelf-life claims at this stage.

Aim 2.3: Administration and preparation robustness
Practical aspects of formulation use will be assessed, including suspension stability, reconstitution behavior (if applicable), dose uniformity, and sensitivity to handling. This sub-aim ensures that optimized formulations are reliable and user-compatible in realistic clinical or home-use scenarios.

Aim 3. Translational Scale-Up and Prototype Demonstration (Suggested timeline: Months 24–36)

The objective of Aim 3 is to establish the translational relevance and manufacturability of optimized formulation systems through pilot-scale development and prototype-level validation.

Aim 3.1: Pilot-scale process development
Scalable manufacturing workflows will be explored for selected formulations, with emphasis on reproducibility, yield, and process robustness. Activities will be designed to inform future GMP translation rather than finalize commercial manufacturing processes.

Aim 3.2: Delivery format compatibility
Optimized formulations will be evaluated for compatibility with clinically relevant delivery formats, such as prefilled syringes or autoinjector systems. Considerations may include short-term storage behavior, material compatibility, and delivery consistency.

Aim 3.3: In vivo feasibility and tolerability assessment
Selected formulations will be assessed in appropriate preclinical models to evaluate pharmacokinetics, absorption behavior, and local tolerability following subcutaneous administration, confirming preservation of biological performance while reducing injection volume and administration burden.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute is expected to play a leading role in **scientific direction and formulation strategy**, while engaging the Korean SME partner throughout the project to incorporate industrial perspectives from an early stage. Anticipated responsibilities can include: • Guiding formulation concept exploration, including defining the formulation design space and exploring candidate formulation architectures in collaboration with the SME partner. • Conducting laboratory-scale formulation and characterization studies, such as protein screening, formulation feasibility assessment, and preliminary injectability and stability evaluations. • Integrating and interpreting experimental data to inform iterative refinement of formulation strategies and prioritization of candidate systems. • Supporting translational evaluation activities, including in vivo feasibility studies, as appropriate, to assess biological performance of selected formulations. • Facilitating technical exchange and coordination, ensuring that feedback from early industrial considerations is reflected in ongoing formulation development.
Expected R&R of Korean SME Partner	The research institute is expected to play a leading role in **scientific direction and formulation strategy**, while engaging the Korean SME partner throughout the project to incorporate industrial perspectives from an early stage. The Korean SME partner is expected to participate as a collaborative contributor, including during early-stage formulation exploration, while bringing practical insights related to materials, processes, and translation. Anticipated responsibilities include: • Early-stage co-development and feasibility input, contributing to formulation design discussions in Aim 1, including selection of protein modalities, formulation architectures, excipients, and carrier systems based on manufacturability and scalability considerations. • Process- and materials-oriented exploration, supporting early formulation studies through material synthesis, excipient modification, pilot-scale screening tools, or process prototyping, as appropriate. • Joint evaluation of injectability and stability, participating in rheological testing, handling assessments, and stress studies to ensure alignment between laboratory performance and industrial feasibility. • Scale-up and manufacturing development, leading pilot-scale production and process optimization of selected candidate formulations identified after Aim 1 and refined in Aim 2. • Delivery format and device compatibility assessment, evaluating formulation performance in clinically relevant delivery systems and contributing to integration strategies.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to establish a foundational formulation platform for injectable ultra-high-concentration (UHC) protein therapeutics, enabling delivery of high-dose biologics in reduced injection volumes suitable for subcutaneous administration. Technical and Scientific Outcomes • Identification and validation of candidate formulation systems (defined by material composition and physical form, e.g., particle-based or hybrid formulations) that enable protein concentrations beyond conventional aqueous limits while maintaining injectability and stability. • Establishment of design principles and structure-property-performance relationships linking protein modality, formulation architecture, excipients, and carrier environments to injectability and robustness. • Demonstration of injectable prototype formulations compatible with clinically relevant needles and delivery formats, with preserved biological performance in preclinical settings. • Generation of experimental data on injectability, handling robustness, and stress stability, supporting future translational and regulatory development. Translational and Industrial Outcomes • Development of pilot-scale-relevant formulation and process concepts that inform manufacturability and scalability without committing to a single commercial configuration. • Early evaluation of delivery system compatibility, such as prefilled syringes or autoinjector formats, to support downstream productization pathways. • Creation of a transferable formulation toolbox that can be applied across multiple protein modalities, disease indications, and dosing requirements. Intellectual Property and Commercialization Potential • Generation of intellectual property related to formulation materials, architectures, and manufacturing approaches arising from co-development activities. • Establishment of technology transfer and commercialization pathways through Korean SMEs, enabling future licensing, scale-up, or product development efforts. • Strengthening of Korean SME competitiveness in advanced biopharmaceutical formulation and manufacturing technologies. Broader Impact and Collaboration Outcomes • Reduction of reliance on intravenous infusion for high-dose biologics, contributing to improved patient convenience, access, and healthcare efficiency. • Support for decentralized and home-based administration of protein therapeutics, with potential impact on global health access. • Strengthened Korea-U.S. (MGB) collaborative R&D ecosystem, fostering long-term partnerships between academic, clinical, and industrial stakeholders. • Dissemination of outcomes through joint publications, presentations, and technical exchanges, contributing to workforce development and international visibility. Commercial and Economic Impact • Potential enablement of next-generation injectable biologic products with reduced administration burden and differentiated market positioning. • Creation of high-value IP and know-how supporting SME growth, technology exports, and participation in global biologics supply chains. • Contribution to the advancement of Korea's Advanced Bio strategic technology field through

	platform-level innovation rather than single-product development.